

Vraj Shah

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Education

Georgia Tech University

Master of Science, Cyber Security

June 2026 - Present

The Ohio State University

Bachelor of Science, Computer Science & Engineering, Columbus, OH

August 2021 - May 2025

Work Experience

Software Engineer II

UST Global Inc., Springfield, OH

June 2025 – Present

- Developing production diagnostic software for heavy-duty vehicles in Java, automating UDS (ISO 14229) and SAE J1939 communication across CAN-based networks to systematically validate and interrogate ECUs.
- Architected core UDS services — Session Control, Security Access, and DTC management — alongside a J1939 diagnostic framework for real-time fault isolation and automated service execution across multiple ECMs.
- Engineered CAN log analysis tooling to parse and decode PGNs, DIDs, and security-relevant traffic, enabling deep inspection of network behavior and surfacing anomalies for cybersecurity review.
- Modernized the diagnostic toolchain with JavaFX and Maven to accelerate in-manufacturing data collection, and standardized J1939 PGN/SPN mappings cross-functionally to reduce downtime and scale to future models.

Frontend Engineer

Hale, Chicago, IL

March 2025 – June 2025

- Led the front-end architecture for Hale's men's care e-commerce platform, building a modular React/Next.js codebase with a shared component library, integrating RESTful micro-service APIs, and surfacing a dynamic product-recommendation UI driven by a weighted-parameter algorithm.

Research Assistant

The Ohio State University, Columbus, OH

August 2024 – May 2025

- Engineered a computer-vision + NLP workflow (Python | WhisperX | Llama-3) that cuts debate-video processing from 78 min to 13 min (6× faster) while auto-tagging speaker, topic, and sentiment.
- Designed a PostgreSQL schema cataloging 10K politicians and 1M timestamped clips, then built matplotlib dashboards that turned that data into insights cited in three journal articles and two conference talks.

Backend Engineer

dustin, Columbus, OH

June 2024 – Dec. 2024

- Architected and deployed "dustin," a revenue-generating AI-powered SaaS for e-commerce analytics, integrating Plaid API for financial aggregation, LangChain for NLP, and Supabase for backend infrastructure — driving customer acquisition through expense tracking and predictive demand forecasting.

Projects and Awards

National Cyber League

Competed in the National Cyber League CTF, applying skills in log analysis, binary exploitation, OSINT, and cryptography — earning a Platinum award (top 100 of 5,380 collegiate competitors nationwide).

October 2018 – Present

Tartan Hacks Hackathon

- Developed a first responder disaster assessment platform utilizing COLMAP for photogrammetric 3D reconstruction, NVIDIA CUDA acceleration, and Next.js front-end framework to identify structural vulnerabilities and locate survivors; won 1st place in the Making Waves track,

February 2025

HackOH/IO Makeathon

Built an ML-powered security suite featuring facial recognition door access, real-time gunshot detection via audio analysis, and Twilio SMS alerting for unknown entries and environmental threats — placing 2nd out of 100 teams.

March 2024

HackOH/IO Hackathon

- Built a Google Maps-powered safety app in 24 hours to identify and display the nearest emergency blue light from a user's location, placing 2nd out of 200 teams.

October 2022

Skills & Certifications

Programming Languages: Java, C++, C, x86-64 Assembly, Python, TypeScript, React Native, JavaScript (ThreeJS, VueJS, NextJS).

Technical Skills: Linux, HTML5, CSS, Git, Microsoft Suite, SQL, Bash, SolidWorks, Visual Studio, Figma, Twilio, WebGL, OpenCV, Tailwind CSS, Mux, SDLs, Rest APIs, WhisperX, NLP, COLMAP, CUDA acceleration, Supabase.

Certifications: Java Foundations (1Z0-811).